INCH-POUND

MIL-DTL-7034/3B
10 March 2022
SUPERSEDING
MIL-DTL-7034/3A
w/Amendment 2
15 February 2012

# DETAIL SPECIFICATION SHEET

# PULLEY, GROOVE, HEAVY DUTY, CONTROL, AIRCRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet, MIL-DTL-7034, and QPL-7034.

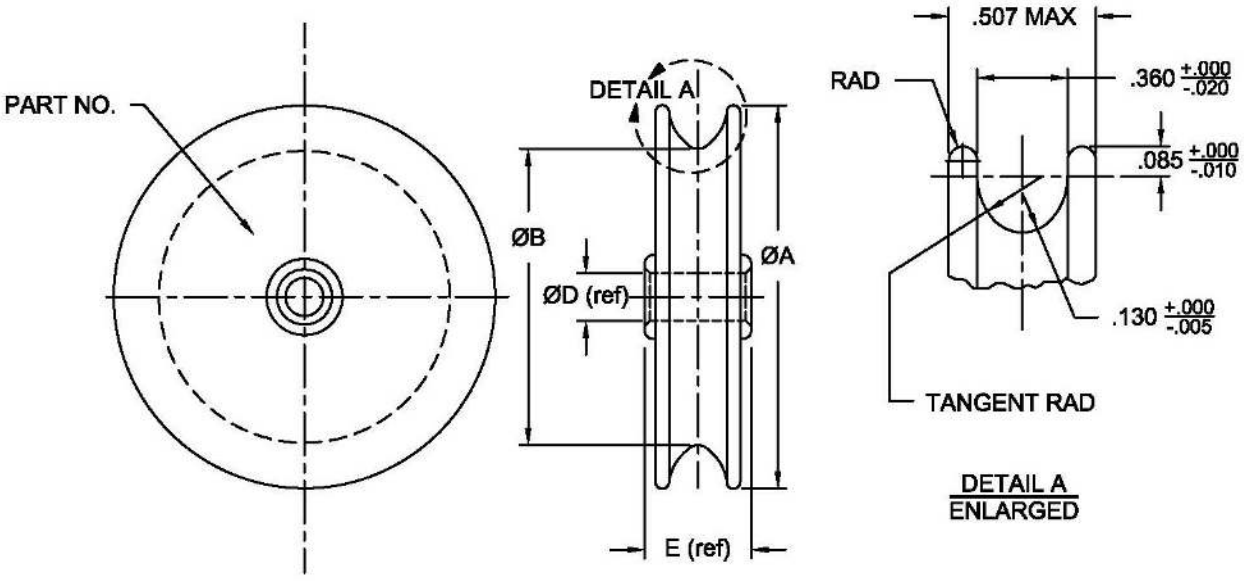



FIGURE 1. Configuration and dimensions.



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TABLE I. Dimensions and interface characteristics.

| Dash no. 1/ | | Cable size (ref) (inches) | A | B | D | E | Pulley assembly max weight (lb) |
|---|---|---|---|---|---|---|---|
| Nonmetallic sheave | Metallic sheave | | Outside dia ±0.005 | Groove dia ±0.005 | Bore dia (ref) 3/ | Width I.R. (ref) 3/ | |
| N/A 2/ | A1 | 3/16 | 2.625 | 2.000 | .31 | .62 | .165 |
| -2 | A2 | 7/32 | 4.125 | 3.500 | .50 | .75 | .450 |
| -3 | A3 | 1/4 | 5.625 | 5.000 | | | .680 |

TABLE I. Dimensions and interface characteristics - Continued.

| Dash no. 1/ | | Pulley strength | | | | SAE AS21443 bearing |
|---|---|---|---|---|---|---|
| Nonmetallic sheave | Metallic sheave | Allowable limit load on pulley (lbs) | Max limit load on cable independent of wrap angle | | | |
| | | | 3/16 cable | 7/32 cable | 1/4 cable | |
| N/A 2/ | A1 | 2,800 | 2,620 lbs | 3,060 lbs | 3,500 lbs | MS21443-5B |
| -2 | A2 | 4,900 | | | | MS21443-8 |
| -3 | A3 | 7,000 | | | | |

1/ See note 2 for example of complete part number.

2/ The -1 nonmetallic pulley was removed from a previous MS standard and applicable specification.

3/ The reference bearing bore diameter and bearing width are controlled by SAE AS21443.

REQUIREMENTS:

1. Material: Material shall be in accordance with MIL-DTL-7034.

2. Finish: Finish shall be in accordance with MIL-DTL-7034.

3. Surface roughness: Surface roughness shall be in accordance with MIL-DTL-7034.

4. Hub may be added to pulley sheave where width of bearing outer race is greater than the maximum width of the pulley sheave.

NOTES:

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

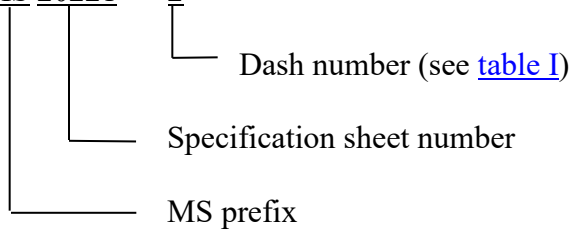
1. Dimensions are in inches. Unless otherwise specified, decimal tolerances: +0.000, -0.010. End faces of all bearing hubs shall be flat and square with bore within ±1°.



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2. The part or identifying number (PIN) to be used for pulleys acquired to this specification is created as shown below. In the transition of MS20221 to detail specification sheet MIL-DTL-7034/3, the MS20221 part numbers have been retained as follows:

EXAMPLE: MS 20221 - 2




MS20221-2 Indicates: Size 2 nonmetallic sheave and fixed ball bearing with contact seal.

MS20221A2 Indicates: Size 2 metallic sheave and fixed ball bearing with contact seal.

All dash numbers in table I are identical to and interchangeable with like dash numbers of MS20221.

3. The replacement sequence for this document is AN221 replaced by MS20221 replaced by MIL-DTL-7034/3.

4. For design feature purposes, this specification sheet takes precedence over the acquisition document referenced herein.

CHANGES FROM PREVIOUS ISSUE. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army – AV
Navy – AS
Air Force – 71
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Preparing Activity:
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(Project 1640-2022-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.dla.mil/.



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